Non-Newtonian Fluid Flow in an Elastic Vessel

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Introduction and background

Anywhere you look you will find fluids. Whether that be in the air which flows around you, the water that you drink, or even the blood flowing through your veins at this very instant. Fluids are everywhere.

Some fluids behave more simply than others. Water, for instance, is quite well behaved: turn a bottle upside down and it comes spilling out. This is an example of a *Newtonian* fluid, a fluid with a constant viscosity (thickness). Other fluids are less well behaved. Ketchup, for example, can stubbornly refuse to leave the bottle when turned upside down! This is an example of a *non-Newtonian* fluid, a fluid whose viscosity (thickness) changes depending on how much force it experiences. Blood is another example, with its thickness changing depending on how quickly it flows.



Figure 1: Examples of different fluids: water, ketchup, and blood.

In this project, we will explore fluid flowing in a vessel. The main application in mind for this project will be blood flowing through a vein or artery. The goal of this project is to produce simulations of non-Newtonian fluids (blood) flowing in a rigid channel, and then extend it to an elastic vessel (veins and arteries), where the walls change shape depending on how the fluid flows through it. To achieve these simulations, we will build on an existing in house program which simulates Newtonian fluids through a rectangular channel.

Reading and understanding the theory

Before starting the project, I knew little about how fluids behave and flow, so I first needed to learn the basics to produce accurate simulations.

Initially, I worked through the fundamental laws of the study of fluids, *fluid mechanics*, the article that we will build on in the project by Anand & Christov and the principles of the numerical method we will use to produce the simulations of the fluid flow, the *Lattice Boltzmann Method* (LBM).

Anand & Christov's article was particularly challenging to work through due to my lack of familiarity with the material going in. In order to overcome this, I took my time, harder than it sounds when you can get carried away with all the interesting maths you are doing, and kept a level head and sieved through it methodically.

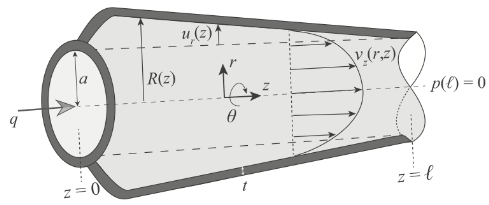

Figure 2: The figure from Anand & Christov article.

The goal of this section was to understand the theory and also be able to implement a key change in the program: to be able to simulate non-Newtonian fluids instead of only Newtonian. After reading through various notes on fluid mechanics, Anand & Christov's article and the basics of the LBM, I felt confident to move on to the next step: understanding the current program and how to modify it.

A personal surprise that came from this section is that I enjoyed learning about the *kinetic theory*, that is the theory that explains the behaviour of gases and fluids from the collective motion and collisions of their tiny constituent particles, much more than I thought I would!

The program and its tweaks

Armed with my newfound fluid knowledge, I began to read through the existing program and figure out what was going on. This part of the project resulted in the largest difficulty but also the most growth.

Initially the 20+ files that made up the program where intimidating and I found it tricky to know where to start. After reading through the main files a few times, I identified and narrowed in on the key steps.

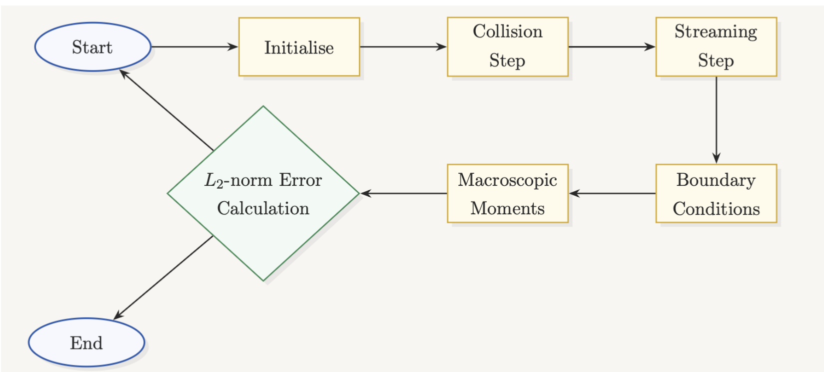

Figure 3: A flowchart of the program.

The program consisted of 6 key steps: initialisation, collision, streaming, boundary conditions, macroscopic moments and an error calculation. If the error was sufficiently small the program would then end or, if the error was too large would repeat. This flowchart single handily saved my sanity.

Once it was clear what the program did, I could then implement a key change that would make the program work for non-Newtonian fluids; this change would take place in the collision step. The change involved adding in the *power-law* which describes how a fluid's viscosity changes with the rate at which it is flowing or being deformed and also doing some theoretical calculations to have something to validate the simulation with. After all was said and done, I finally had some pretty pictures to show for it.

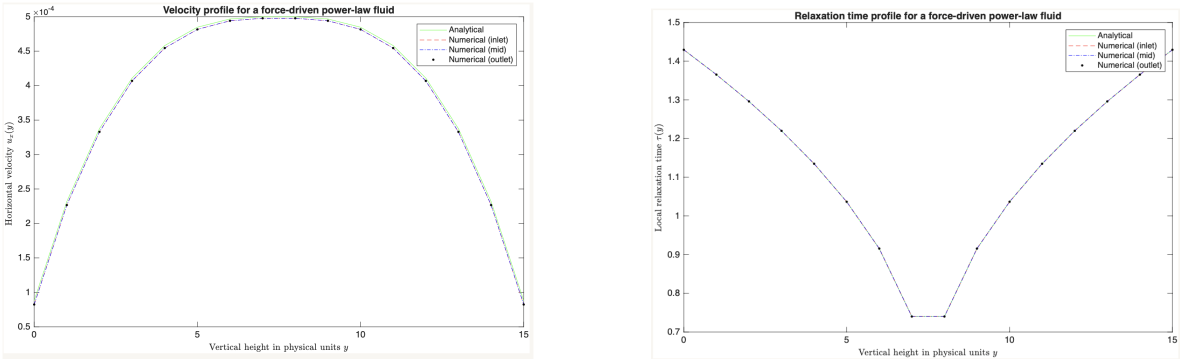

Figure 4: Results from the implementation.

Mesh convergence studies

Now that we had a working program that simulated non-Newtonian fluids in a rigid channel, we had to now validate it even further with a *mesh convergence study.* This study involves running multiple simulations with larger and larger *meshes*, the space in which the simulations take place, and with longer and longer iteration times.

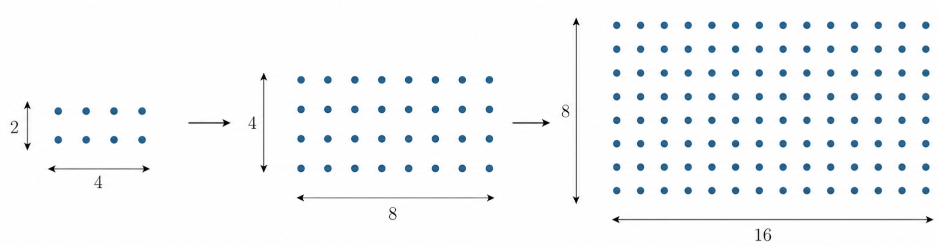

Figure 5: Diagram showing the mesh size scaling for the mesh convergence study.

An example of this procedure is as follows. Pick a starting mesh size of 2 by 4, and then run it for 100 iterations. For each subsequent mesh, double the previous length and width and quadruple the number of iterations. The next mesh would be 4 by 8 and would be run for 400 iterations and so on.

This mesh convergence study was to show that our program was converging as expected by looking at the errors of each simulation. Each simulation has a margin of error to the expected value. This error *should* half for each subsequent mesh. Thankfully, after running these meshes, that is exactly what we saw and we can now move on to extending our simulations to elastic vessels.

Extension to elastic vessels

Now that we have all the pieces together, we can turn on the elastic walls and see what happens! At each stage so far, there was a theoretical solution that we could compare our simulation to so that we know it was on the right track. Now, there is no theoretical solution to a non-Newtonian fluid flowing through a elastic vessel, so what we would see from our simulation would be new information!

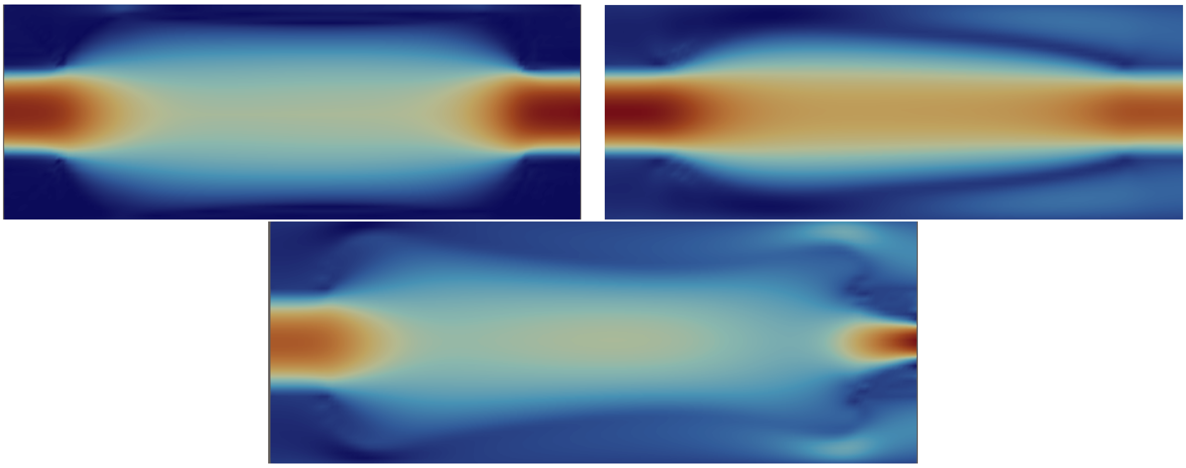

Figure 6: Various results of our extension to an elastic vessel.

The images above show the velocity profile within the vessel, with the velocity magnitude being its highest in the regions shown in red and decreases progressively through orange, yellow, and green, reaching its lowest values in the regions shown in blue. You can also see the deformation of the vessel, particularly in the lower image. If we had some more time, the next step would have been to conduct a mesh convergence study for this new elastic vessel.

Conclusion

After working through the theory, the C++ program, mesh convergence studies and turning on the elastic vessels, we achieved our goal of producing non-Newtonian fluid simulations in a flexible vessel which can then aid in modelling blood flow in the human body.

The experience of working on serious and eye opening mathematical research has been extraordinary and has inspired me to continue to pursue a career in mathematical research. In addition, the skills of mathematical modelling, C++ programming and numerical methods gained in this project will prove to be invaluable.

Acknowledgements

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